

Open-query games on ordinal spaces

A negative solution to the topological-sum problem,
with an exact Cantor–Bendixson classification

Research report

19 September 2026

For the convergent ordinal sequence $S = [0, \omega]$,

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2.$$

More generally, for every integer $k \geq 1$, the compact ordinal space

$$K_k = [0, \omega^{2^k - 1}]$$

satisfies $\text{sm}(K_k) = k$ and $\text{sm}(K_k \sqcup K_k) = k + 1$.

Complete ordinary mathematical proofs are included. The archive also contains an independent finite-poset verifier, exact data, and a machine-readable strategy certificate. The infinite-space results do not depend on numerical experiments. No claim of independent peer review or formal proof-assistant verification is made.

Abstract

We answer negatively Problem 1.3 of Chiozini, Csernák, and Soukup’s *Gamification of the T_0 -pseudoweight via cut-and-choose games on topological spaces*, in the version available as arXiv:2510.05754v3. The counterexample consists of two convergent sequences, so it lies entirely within compact metrizable ordinal spaces.

The main further result is an exact formula for every nonempty Hausdorff space of finite Cantor–Bendixson height. If $X^{(h)} \neq \emptyset$ and $X^{(h+1)} = \emptyset$, then

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases}$$

The proof identifies finite open-query complexity with the logarithm of the minimum length of an alternating closed-layer decomposition. A canonical alternating-closure procedure computes that length and retains the orientation information lost by the scalar invariant sm .

Consequences include sharp one-move failures of the proposed sum identity at every positive finite level, exact finite-product formulae, a classification on countable scattered Hausdorff spaces, and a quadratic-time target algorithm for finite posets. Independent minimax computations verify the finite normal form on all naturally labelled posets through six elements and all their targets: 320,866 tests in total.

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1 The selected problem and the status of the answer

In the set-membership game, questions are restricted to open subsets of a space, and the aim is to decide membership in a fixed target subset. The invariant $\text{sm}(X)$ is the worst, over all targets, of the least ordinal number of questions needed. Chiozini, Csernák, and Soukup ask whether

$$\text{sm}\left(\bigsqcup_{i \in I} X_i\right) \stackrel{?}{=} \sup_{i \in I} \text{sm}(X_i). \quad (1.1)$$

This is their Problem 1.3; their Theorem 3.4 gives the upper bound consisting of the right-hand side followed by one additional move [1].

The answer to (1.1) is *no*. The proof in the next section is short and does not depend on any later theorem. In particular, the counterexample is not an artifact of non-Hausdorff topology, uncountable spaces, or an ambiguity about transfinite play. It uses a two-move game on a countable compact metrizable space. Repeating the same component countably many times also gives a counterexample when the index set is required to be infinite.

The more substantial development is the classification in theorem 6.2. Its mechanism explains exactly why the counterexample works. Different components may require opposite interpretations of the same open-set answer. A scalar question count forgets that orientation, while a pair of alternating-closure lengths preserves it.

Status and provenance. The arXiv record retrieved for this report lists version 3, dated 29 May 2026; the question remains explicitly printed there [1]. The report gives a negative solution to that stated question, not a solution to every open problem in the paper. A targeted literature search found no intervening resolution of this question. This is a search statement, not a proof of priority. The closed-layer machinery belongs to the classical setting of Hausdorff difference hierarchies; we do not claim that framework as new [2]. All lemmas needed for the results here are proved below.

2 A compact ordinal counterexample

Let $S = [0, \omega]$ have its order topology. Thus every natural number is isolated, and the neighborhoods of ω contain a final segment of $\mathbb{N}$. We use zero questions for a constant target: a subset equal to the whole space or to the empty set.

Lemma 2.1 (The one-question criterion). *For a subset A of any topological space X , membership in A can be decided with at most one open-set question if and only if A is open or closed.*

Proof. An open target is itself a legal question. A closed target can be tested by asking its open complement. Conversely, suppose the first question is U . The only two possible final states are U and $X \setminus U$. Both must be monochromatic for membership in A . If they have the same color, A is constant. Otherwise $A = U$ or $A = X \setminus U$. $\square$

Proposition 2.2. *For $S = [0, \omega]$,*

$$\text{sm}(S) = 1, \quad \text{sm}(S \sqcup S) = 2. \quad (2.1)$$

Proof. Every subset of S missing ω is open, since it consists entirely of isolated points. Every subset containing ω is closed, since its complement consists of isolated points. Hence every target takes at most one question. There are nonconstant targets, so $\text{sm}(S) = 1$.

Write S_0, S_1 for two disjoint copies, with limit points p_0, p_1 , and put

$$A = \{p_0\} \sqcup (S_1 \setminus \{p_1\}).$$

This set is not open: it contains p_0 but no isolated point of S_0 . It is not closed: it contains all isolated points of S_1 , but omits their limit p_1 . By lemma 2.1, one question cannot suffice.

Two questions suffice for *every* target B . First ask the clopen set S_0 , thereby determining which copy contains the remaining candidates. Inside that copy, B is open or closed, so one more open question decides membership. An open set in a component is open in the disjoint union. $\square$

The fact that spaces with at most one accumulation point have one-question membership complexity is also noted in [1, Section 7]; the failure of (1.1) comes from putting two such components together.

Corollary 2.3. *The proposed identity fails even when every component and the whole sum are compact, countable, metrizable, zero-dimensional, and homeomorphic to ordinal spaces. Specifically,*

$$S \sqcup S \cong [0, \omega \cdot 2], \quad \text{sm}([0, \omega \cdot 2]) = 2.$$

Proof. The two clopen intervals $[0, \omega]$ and $(\omega, \omega \cdot 2]$ partition $[0, \omega \cdot 2]$. Each has order type $\omega + 1$. The stated topological properties can also be seen directly from two disjoint convergent sequences. $\square$

Proposition 2.4 (An infinite index set). *For every cardinal $\kappa \geq 2$,*

$$\text{sm}\left(\bigsqcup_{i < \kappa} S\right) = 2.$$

Proof. The lower bound follows by restricting to two components. For the upper bound, fix a target A , and split the component indices into two classes so that A_i is open on the first class and closed on the second; when both apply, choose either class. One question asks the union of the components in the first class. A second question asks the open set whose trace is A_i on the first class and $S_i \setminus A_i$ on the second. The two answers determine the color. $\square$

3 The game, subspaces, and ordinal conventions

All ordinal intervals in this report carry their order topology. The notation $[0, \gamma]$ denotes a space whose underlying ordered set has order type $\gamma + 1$. A topological sum $X \sqcup Y$ is not being confused with an ordinal sum; when they are homeomorphic, this is stated separately.

Definition 3.1 (Open-query membership game). Fix $A \subseteq X$ and an ordinal β . At stage $\xi < \beta$, the Seeker chooses an open set U_ξ . The Hider answers $e_\xi \in \{0, 1\}$. Set

$$W_\eta = \bigcap_{\xi < \eta} \begin{cases} U_\xi, & e_\xi = 1, \\ X \setminus U_\xi, & e_\xi = 0. \end{cases}$$

The Seeker wins if $W_\beta \subseteq A$ or $W_\beta \cap A = \emptyset$. Write $\text{sm}(A, X)$ for the least winning length and $\text{sm}(X) = \sup_{A \subseteq X} \text{sm}(A, X)$, as in [1].

For ordinal-valued invariants we work with T_0 spaces, where a winning length exists: asking all open sets separates points. A statement of the form $\text{sm}(A, X) \leq d$ can also be read on an arbitrary topological space as the existence of a winning depth- d strategy, whether or not a least ordinal winning length exists. The finite normal form is valid in this broader sense.

An empty final state is winning. The Hider need not commit to a point in advance. For a finite decision tree, however, the game formulation and correctness on every fixed point are equivalent: a bad leaf contains two points of different colors, while an empty leaf is harmless. This equivalence will be used carefully in section 4.

Lemma 3.2 (Restriction to a subspace). *If $Z \subseteq X$ and $A \subseteq X$, then*

$$\text{sm}(A \cap Z, Z) \leq \text{sm}(A, X), \quad \text{sm}(Z) \leq \text{sm}(X).$$

Proof. Restrict every open question in a strategy on X to Z . It is open in the subspace, and every resulting state is the old state intersected with Z . For the second inequality, extend an arbitrary target of Z to a target of X . $\square$

Finite question counts are ordinary nonnegative integers. We use $\lceil \log_2 L \rceil$ only for positive integers L ; equivalently, it is the least integer d for which $L \leq 2^d$. This avoids floating-point or ordinal-logarithm conventions. In particular, $\lceil \log_2 1 \rceil = 0$.

4 Finite decision trees and alternating layers

Definition 4.1 (Alternating closed layers). For a nonempty space X and $A \subseteq X$, let $C_1 = A$ and $C_0 = X \setminus A$. A closed-layer decomposition of length $L \geq 1$ and outer color $c \in \{0, 1\}$ is a chain

$$X = F_0 \supseteq F_1 \supseteq \cdots \supseteq F_L = \emptyset \tag{4.1}$$

of closed sets such that

$$F_i \setminus F_{i+1} \subseteq C_{(c+i) \bmod 2} \quad (0 \leq i < L). \tag{4.2}$$

Empty layers are allowed. Let $L(A, X)$ be the least such length, allowing either outer color, and put $L(A, X) = \infty$ if no finite chain exists.

These are finite alternating differences of closed sets, equivalently finite Boolean combinations of open sets. This connects the game to the difference hierarchy, but the exact decision-depth statement below is proved directly.

Theorem 4.2 (Finite normal form). *For every nonempty topological space X , target A , and integer $d \geq 0$,*

$$\text{sm}(A, X) \leq d \iff L(A, X) \leq 2^d. \tag{4.3}$$

Consequently, whenever $L(A, X) < \infty$,

$$\text{sm}(A, X) = \lceil \log_2 L(A, X) \rceil. \tag{4.4}$$

4.1 From a query tree to an ordered collection of leaves

Lemma 4.3. *Complete a finite open-query tree to depth d , padding terminal branches with dummy questions. Number its leaves $0, 1, \dots, 2^d - 1$ in lexicographic order of answer strings, with $0 < 1$. If $\ell(x)$ is the leaf reached by the answers of x , then every upper level set*

$$\{x : \ell(x) \geq j\}$$

over the leaf order is open.

Proof. Proceed by induction on d . The case $d = 0$ is immediate. Let the root question be U , and put $M = 2^{d-1}$. Consider the two subtrees as query trees defined on all of X , regardless of which root answer would actually occur, and write ℓ_0, ℓ_1 for their leaf functions. If $j \leq M$, then

$$\{\ell \geq j\} = U \cup \{\ell_0 \geq j\};$$

for $j = M$, the second set is empty. If $j > M$, then

$$\{\ell \geq j\} = U \cap \{\ell_1 \geq j - M\}.$$

The induction hypothesis and closure of the topology under finite unions and intersections prove the claim. $\square$

Proof of the forward implication of theorem 4.2. A winning depth- d tree has a well-defined output color at each nonempty leaf. Assign an arbitrary color to each empty leaf. Read the 2^d leaf colors from the largest leaf down to the smallest, and group consecutive equal colors into runs. There are at most 2^d runs, and successive run colors alternate.

After deleting the largest i runs, the set of points still represented is a lower level set of ℓ . It is closed by lemma 4.3. These lower level sets, starting with X and ending with $\emptyset$, form the required chain. A run can represent no points, which is why empty layers were allowed. $\square$

4.2 Balanced search among closed layers

Proof of the reverse implication of theorem 4.2. Suppose (4.1)–(4.2) hold. Every point lies in a unique layer $F_i \setminus F_{i+1}$. The open question $X \setminus F_j$ tests whether $i < j$. Thus ordinary binary search can identify the layer index among L possibilities. Split the current interval of indices as evenly as possible at each question. Its largest remaining size after d questions is at most $\lceil L/2^d \rceil$. When $L \leq 2^d$, at most one layer remains, and that layer is monochromatic. The same reasoning applies to arbitrary Hider answers: either the state is empty or it is contained in a single layer. $\square$

Remark 4.4 (Why the logarithm is genuine). Sequentially peeling L layers would spend roughly L questions. Adaptivity permits balanced search through nested open thresholds, reducing the count to $\lceil \log_2 L \rceil$. Conversely, a depth- d tree has at most 2^d ordered leaf runs. Neither direction assumes that the space is ordered or Hausdorff.

5 Canonical alternating closures and orientation

The least layer length has a concrete canonical description. This is also the right quantity for understanding disjoint unions.

Definition 5.1 (Oriented closure sequence). For $c \in \{0, 1\}$ define

$$F_0^c = X, \quad F_{i+1}^c = \text{cl}_X(F_i^c \cap C_{(c+i+1) \bmod 2}). \quad (5.1)$$

Let $N_c(A, X)$ be the first integer L such that $F_L^c = \emptyset$, or ∞ when no such integer exists. The pair (N_0, N_1) is the *oriented closure profile* of the target.

Every F_i^c is closed and the sequence is decreasing. At step $i + 1$, one keeps the closure of points whose color would be wrong in the next layer. Therefore

$$F_i^c \setminus F_{i+1}^c \subseteq C_{(c+i) \bmod 2}.$$

Theorem 5.2 (Canonical minimality). *For each outer color c , $N_c(A, X)$ is the least possible length of a closed-layer decomposition with that color. In particular,*

$$L(A, X) = \min(N_0(A, X), N_1(A, X)). \quad (5.2)$$

Proof. If the canonical sequence terminates, its layers are a decomposition of the required length. Conversely, let $G_0, \dots, G_L$ be any closed-layer chain with outer color c . We show $F_i^c \subseteq G_i$ by induction. The assertion is true at $i = 0$. If it holds at i , then every wrong-colored point of F_i^c belongs to G_{i+1} , because $G_i \setminus G_{i+1}$ has the prescribed color. Taking closures preserves containment in the closed set G_{i+1} . Hence $F_{i+1}^c \subseteq G_{i+1}$. In particular $F_L^c = \emptyset$. $\square$

Lemma 5.3 (Two elementary profile rules). *Complementing the target swaps N_0 and N_1 . Also,*

$$N_c \leq N_{1-c} + 1$$

whenever the latter is finite.

Proof. The first assertion follows directly from (5.1). For the second, prepend an empty outer layer of color c to a decomposition whose outer color is $1 - c$; in the chain this means repeating X once at the beginning. Apply theorem 5.2. $\square$

Thus finite profile entries differ by at most one, but discarding the distinction between them can change a question count at a power-of-two threshold.

Proposition 5.4 (Exact finite profile of a sum). *Let $X = \bigsqcup_{i \in I} X_i$, $A_i = A \cap X_i$, and let $c \in \{0, 1\}$. For a nonempty family of nonempty components,*

$$N_c(A, X) = \sup_{i \in I} N_c(A_i, X_i), \quad (5.3)$$

where the supremum is taken in $\mathbb{N} \cup \{\infty\}$: an unbounded set of finite lengths has value ∞ . Hence, whenever the resulting minimum is finite,

$$\text{sm}(A, X) = \left\lceil \log_2 \left(\min_{c \in \{0, 1\}} \sup_i N_c(A_i, X_i) \right) \right\rceil. \quad (5.4)$$

Proof. Closure in a disjoint union is computed componentwise. Inductively, $F_n^c(A, X) \cap X_i = F_n^c(A_i, X_i)$ for every finite n . The global sequence is empty exactly when every component sequence is empty. Now use theorems 4.2 and 5.2. $\square$

The obstruction to (1.1) is already visible in the elementary inequality

$$\min_c \sup_i N_{i,c} \geq \sup_i \min_c N_{i,c}.$$

The two operations need not commute: each component may have its own preferred outer color, whereas one global closed-layer chain has only one outer color.

6 Exact classification at finite Cantor–Bendixson height

For a closed subspace F of a Hausdorff space, let F' be its set of nonisolated points. Set $X^{(0)} = X$, $X^{(n+1)} = (X^{(n)})'$; at a limit ordinal, use the intersection of the previous derivatives. Only finite derivatives are needed for the principal formula.

Definition 6.1. A nonempty Hausdorff space has *last derivative rank* h if

$$X^{(h)} \neq \emptyset, \quad X^{(h+1)} = \emptyset.$$

Its Cantor–Bendixson height is then $h + 1$. The terminal derivative $X^{(h)}$ is discrete; it need not be finite unless the space is compact.

Theorem 6.2 (Finite-height classification). *Let X be a nonempty Hausdorff space with last derivative rank $h < \omega$. Then*

$$\text{sm}(X) = \begin{cases} \lceil \log_2(h+1) \rceil, & |X^{(h)}| = 1, \\ \lceil \log_2(h+2) \rceil, & |X^{(h)}| \geq 2. \end{cases} \quad (6.1)$$

More precisely, the maximum of $L(A, X)$ over all targets is $h + 1$ in the first case and $h + 2$ in the second. The maxima are attained.

6.1 Why canonical closures lose one derivative at each step

Lemma 6.3. *If $F = \text{cl}_X B$ and x is isolated in F , then $x \in B$.*

Proof. An open neighborhood U with $U \cap F = \{x\}$ must meet B , because $x \in \text{cl}_X B$. Thus $x \in B$. $\square$

Lemma 6.4 (Derivative descent). *For every target and outer color, the canonical sequence satisfies*

$$F_{n+1}^c \subseteq (F_n^c)' \quad (n \geq 1), \quad F_n^c \subseteq X^{(n-1)} \quad (n \geq 1). \quad (6.2)$$

Proof. By lemma 6.3, every isolated point of F_n^c has color $(c + n) \bmod 2$. Therefore the opposite-colored points whose closure defines F_{n+1}^c all lie in $(F_n^c)'$. This derivative is closed in X , so their closure lies there too. Derivation is monotone on closed sets, giving the second assertion by induction from $F_1^c \subseteq X$. $\square$

Upper bounds in theorem 6.2. For every target and orientation, lemma 6.4 gives $F_{h+2}^c \subseteq X^{(h+1)} = \emptyset$. Hence $L(A, X) \leq h + 2$.

Now assume $X^{(h)} = \{p\}$, and write d for the target color of p . Choose c so that $(c + h) \bmod 2 = d$. We have $F_{h+1}^c \subseteq \{p\}$. If this set were nonempty, p would be isolated in it and, by lemma 6.3, would have color $(c + h + 1) \bmod 2 = 1 - d$, a contradiction. Thus $F_{h+1}^c = \emptyset$ and $L(A, X) \leq h + 1$. $\square$

6.2 Alternating rank layers give matching lower bounds

Write

$$D_j = X^{(j)} \setminus X^{(j+1)} \quad (0 \leq j \leq h).$$

These sets partition X . The set D_j is dense in $X^{(j)}$. Here are details that do not assume compactness: a finite-height space is scattered, because any nonempty subspace contains a point of least ambient rank, and that point is isolated in the subspace after restricting an isolating neighborhood at its rank. Every nonempty open subset of a scattered space contains a point isolated in that space. Apply this to $X^{(j)}$.

Lemma 6.5 (Parity target). *Color every point of D_j by $(b + j) \bmod 2$, where $b \in \{0, 1\}$, and let A_b be the color-1 set. Then*

$$N_b(A_b, X) = h + 1, \quad N_{1-b}(A_b, X) = h + 2. \quad (6.3)$$

Proof. For $h = 0$, the target is constant of color b , and the two lengths are 1 and 2, as asserted. Assume $h \geq 1$. With outer color b , the first closure is X' , because the opposite color contains D_1 , which is dense in X' , and is contained in X' . Repeating the argument shows $F_n^b = X^{(n)}$ for $0 \leq n \leq h + 1$. Thus the first empty term has index $h + 1$.

With outer color $1 - b$, the first closure is X , since it includes the dense rank-zero layer D_0 . Subsequent closures give $F_n^{1-b} = X^{(n-1)}$ for $1 \leq n \leq h + 2$. The first empty term now has index $h + 2$. The same computation covers $h = 0$. $\square$

Lemma 6.6 (Restriction of profiles). *For any subspace $Z \subseteq X$ and either color c ,*

$$N_c(A \cap Z, Z) \leq N_c(A, X).$$

Proof. The closure identity $\text{cl}_Z(B \cap Z) \subseteq Z \cap \text{cl}_X B$ gives, inductively, $F_n^c(A \cap Z, Z) \subseteq Z \cap F_n^c(A, X)$. $\square$

Lower bounds in theorem 6.2. The parity target of lemma 6.5 always has $L = h + 1$, proving the required lower bound when the terminal derivative is a singleton.

Suppose instead that $p \neq q$ lie in $X^{(h)}$. Take disjoint open neighborhoods U, V of p, q . Derivatives commute with restriction to an open subspace: $U^{(j)} = U \cap X^{(j)}$, and similarly for V . In particular both U and V have last derivative rank h .

On U , choose the parity target with bottom color 0. On V , choose the parity target with bottom color 1. Define the target arbitrarily outside $U \cup V$. By lemma 6.5, orientation 0 requires $h + 2$ layers on V , while orientation 1 requires $h + 2$ layers on U . By lemma 6.6, both global orientations require at least $h + 2$ layers. The upper bound already proved gives equality. Applying theorem 4.2 finishes the proof. $\square$

Remark 6.7 (What information survives). At finite height, the invariant sees the last derivative rank and whether the terminal derivative has one point or more than one. It sees neither the exact larger cardinality of that derivative nor the cardinalities of lower levels. Compactness and metrizability are not needed for theorem 6.2.

7 Exact values on ordinal intervals

For $h, m \geq 1$ finite, put

$$X_{h,m} = [0, \omega^h \cdot m].$$

These are countable compact ordinal spaces. Standard background on ordinal spaces and Cantor–Bendixson characteristics is discussed in [3]; the specific derivative calculation needed here is elementary.

Lemma 7.1. *For $1 \leq j \leq h$,*

$$X_{h,m}^{(j)} = \{\omega^j \eta : 0 < \eta \leq \omega^{h-j} \cdot m\}. \quad (7.1)$$

In particular,

$$X_{h,m}^{(h)} = \{\omega^h, \omega^h \cdot 2, \dots, \omega^h \cdot m\}, \quad X_{h,m}^{(h+1)} = \emptyset.$$

Proof. The nonisolated points of an ordinal interval are exactly its nonzero limit ordinals. Below $\omega^h \cdot m$, these are precisely the positive multiples of ω . This gives the formula for $j = 1$.

Within the ordered set of positive multiples of ω^j , a point with successor coefficient is isolated, including the first coefficient. A point with limit coefficient is a limit of smaller such multiples. The nonzero limit coefficients are the multiples of ω . Therefore the next derivative consists exactly of the positive multiples of ω^{j+1} in the indicated range. This proves the induction. At step h the remaining finite set is discrete, so the next derivative is empty. $\square$

Corollary 7.2 (Finite-exponent ordinal formula). *For positive finite h, m ,*

$$\text{sm}([0, \omega^h \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & m = 1, \\ \lceil \log_2(h+2) \rceil, & m \geq 2. \end{cases} \quad (7.2)$$

Last rank h	One terminal point	At least two	Duplication increases sm
0	0	1	yes
1	1	2	yes
2	2	2	no
3	2	3	yes
4	3	3	no
5	3	3	no
6	3	3	no
7	3	4	yes
8	4	4	no
15	4	5	yes
31	5	6	yes
63	6	7	yes

The row $h = 0$ refers to arbitrary discrete spaces, not the notation $[0, \omega^0 \cdot m]$. A singleton has value 0; any larger discrete space has value 1.

Explicit hard targets. On $X_{h,1}$, color a point according to the parity of its Cantor–Bendixson rank. In Cantor normal form, the rank of a nonzero point is its last nonzero exponent; the point 0 has rank 0. On two copies, reverse all colors in one copy. The canonical closure lengths are then $h + 2$ in both orientations. This is an explicit witness, not merely an existence argument using the supremum over targets.

Why two copies form another ordinal space. For $h \geq 1$, the intervals

$$[0, \omega^h], \quad (\omega^h, \omega^h \cdot 2]$$

are clopen in $[0, \omega^h \cdot 2]$ and are each homeomorphic to $[0, \omega^h]$. The second interval has the same order type because removing the first point from an infinite indecomposable ordinal ω^h does not change that order type. Thus

$$X_{h,1} \sqcup X_{h,1} \cong X_{h,2}.$$

8 Sharp failures at every finite level, and sums more generally

Theorem 8.1 (Sharpness at every positive finite level). *For every integer $k \geq 1$, let*

$$K_k = [0, \omega^{2^k-1}].$$

Then

$$\text{sm}(K_k) = k, \quad \text{sm}\left(\bigsqcup_{i < \kappa} K_k\right) = k + 1 \quad \text{for every cardinal } \kappa \geq 2. \quad (8.1)$$

Thus the one-extra-move upper bound for sums is attained at every positive finite level, including by sums of two compact metrizable ordinal spaces.

Proof. Put $h = 2^k - 1$. Derivatives of a disjoint union are computed componentwise. Each component has last rank h and one terminal point; a sum of at least two has the same last rank and at least two terminal points. By theorem 6.2, the respective values are

$$\lceil \log_2(2^k) \rceil = k, \quad \lceil \log_2(2^k + 1) \rceil = k + 1.$$

□

Corollary 8.2 (Exact duplication threshold). *If X is Hausdorff of last finite derivative rank h and $|X^{(h)}| = 1$, then*

$$\text{sm}(X \sqcup X) > \text{sm}(X) \iff h + 1 \text{ is a power of 2.}$$

In that case the increase is exactly one.

Proof. The comparison is between $\lceil \log_2(h + 1) \rceil$ and $\lceil \log_2(h + 2) \rceil$. Consecutive positive integers cross a ceiling-logarithm threshold exactly when the smaller is a power of two. □

8.1 Bounded finite heights

Let $\{X_i : i \in I\}$ be any nonempty family of nonempty Hausdorff spaces with finite last derivative ranks h_i . Suppose $H = \sup_i h_i < \omega$. Then H is attained and

$$\left(\bigsqcup_i X_i \right)^{(H)} = \bigsqcup_{h_i=H} X_i^{(H)}.$$

If there is a unique component of height $H + 1$ and that component has a singleton terminal derivative, the sum has value $\lceil \log_2(H + 1) \rceil$. In all other cases its value is $\lceil \log_2(H + 2) \rceil$. These two formulae can give the same numerical value.

8.2 Unbounded finite heights

The next upper bound is useful because the number of components need not be countable.

Proposition 8.3 (Parallel play with finite-time certification). *If every component X_i has finite $\text{sm}(X_i)$, then*

$$\text{sm}\left(\bigsqcup_i X_i\right) \leq \omega.$$

If the component values are unbounded among the integers, equality holds.

Proof. Fix a target A in the sum and, for each i , a finite winning strategy for $A_i = A \cap X_i$. We describe countably many rounds, each using three questions.

In round n , first ask the union of the next local questions in every component whose local strategy has not finished. Use a dummy open question in components that have finished. The single global answer advances all unfinished local simulations along the same answer history.

Next ask the union of the whole components whose current local simulation has finished with output color 0. Finally ask the analogous union for color 1. These are open sets because whole components are open. At either of these certification questions, a yes answer leaves a monochromatic state, and the Seeker has won. Empty local states may be assigned either output color.

Suppose all certification answers are no. Any point surviving the final intersection would lie in a particular component X_i . Its local strategy finishes after finitely many rounds. In that round or an earlier one, a certification question includes the whole component X_i , and a no answer removes the point. This is a contradiction. Thus the final state is empty.

The three questions per round have total order type $3 \cdot \omega = \omega$. For the lower bound when local values are unbounded, use subspace monotonicity. $\square$

In particular, for any family of finite-height Hausdorff spaces whose last ranks are unbounded in ω , the sum has membership number exactly ω . This conclusion is independent of the cardinality of the index set.

9 Finite products and the information lost by sm

Lemma 9.1 (Product derivatives). *Let X, Y be Hausdorff spaces of finite last derivative ranks h, k . For finite n ,*

$$(X \times Y)^{(n)} = \bigcup_{i+j=n} X^{(i)} \times Y^{(j)}. \quad (9.1)$$

Thus the last rank of the product is $h + k$, and its terminal derivative is $X^{(h)} \times Y^{(k)}$.

Proof. For closed subspaces $F \subseteq X$, $G \subseteq Y$, a point of $F \times G$ is isolated precisely when both coordinates are isolated in their respective subspaces. Therefore

$$(F \times G)' = (F' \times G) \cup (F \times G').$$

Derivation distributes over finite unions of closed sets: if a point is isolated or absent in every one of finitely many closed sets, intersect finitely many suitable neighborhoods to witness that it is not an accumulation point of their union. The reverse inclusion is immediate. Induction proves (9.1). At $n = h + k$ only the term with $i = h, j = k$ can be nonempty, and all terms vanish at the next stage. $\square$

Corollary 9.2 (Exact product formula). *For nonempty finite-height Hausdorff spaces $X_1, \dots, X_r$, put $H = \sum_j h_j$. Then*

$$\text{sm}\left(\prod_{j=1}^r X_j\right) = \begin{cases} \lceil \log_2(H+1) \rceil, & \text{every } X_j^{(h_j)} \text{ is a singleton,} \\ \lceil \log_2(H+2) \rceil, & \text{otherwise.} \end{cases}$$

In particular, for finite $r \geq 1$,

$$\text{sm}([0, \omega]^r) = \lceil \log_2(r+1) \rceil. \quad (9.2)$$

Knowing only the values of sm is not sufficient to compute product values. For example, let

$$P = [0, \omega^2], \quad Q = [0, \omega^3], \quad S = [0, \omega].$$

Then $\text{sm}(P) = \text{sm}(Q) = 2$, but

$$\text{sm}(P \times S) = 2, \quad \text{sm}(Q \times S) = 3.$$

The finer data consisting of the last rank and singleton-versus-multiple terminal derivative are compositional for finite products and bounded-height sums. The logarithmic compression to sm discards precisely the numerical information needed in this example.

10 Countable spaces and the transfinite-height boundary

The finite formula is not a transfinite ordinal-logarithm formula. On countable scattered spaces, all infinite Cantor–Bendixson heights lead to the same game value, namely ω .

The point-separating number $\text{ps}(X)$ is defined by the same game, but with winning condition $|W_\beta| \leq 1$ instead of monochromaticity for a target $[1]$.

Lemma 10.1 (A countable T_1 upper bound). *If X is countable and T_1 , then $\text{sm}(X) \leq \omega$. If X is also infinite, then $\text{ps}(X) = \omega$.*

Proof. Enumerate $X = \{x_0, x_1, \dots\}$. Ask the open complements $X \setminus \{x_n\}$ successively. A no answer leaves at most the singleton $\{x_n\}$. If every answer is yes, the final state is empty. This proves both upper bounds. A finite binary tree has finitely many leaves and cannot separate all points of an infinite space, proving the point-separation lower bound. $\square$

Theorem 10.2 (Countable scattered classification). *Let X be a nonempty countable scattered Hausdorff space. If its Cantor–Bendixson height is finite, theorem 6.2 applies. Otherwise*

$$\text{sm}(X) = \omega. \quad (10.1)$$

Proof. Only the lower bound in the infinite-height case remains. For every finite h , $X^{(h)}$ is nonempty and scattered, hence has an isolated point p . Choose an open neighborhood U in X with $U \cap X^{(h)} = \{p\}$. Derivatives commute with open restriction, so

$$U^{(h)} = \{p\}, \quad U^{(h+1)} = \emptyset.$$

By theorem 6.2 and subspace monotonicity, $\text{sm}(X) \geq \lceil \log_2(h+1) \rceil$. These integers are unbounded as h varies. Together with lemma 10.1, this yields (10.1). $\square$

Corollary 10.3 (All countable exponent intervals). *Let $0 < \alpha < \omega_1$ and $1 \leq m < \omega$. Then*

$$\text{sm}([0, \omega^\alpha \cdot m]) = \begin{cases} \lceil \log_2(h+1) \rceil, & \alpha = h < \omega, \ m = 1, \\ \lceil \log_2(h+2) \rceil, & \alpha = h < \omega, \ m \geq 2, \\ \omega, & \alpha \geq \omega. \end{cases}$$

Every one of these spaces has $\text{ps} = \omega$.

Proof. The finite-exponent cases were proved in corollary 7.2. When $\alpha \geq \omega$, the interval contains $[0, \omega^h]$ as a clopen subspace for every positive finite h . Their membership numbers are unbounded, while the whole space is countable and T_1 . Use lemma 10.1. $\square$

For example,

$$\text{sm}([0, \omega^\omega]) = \text{sm}([0, \omega^{\omega^2}]) = \text{sm}([0, \omega^{\omega^\omega}]) = \omega.$$

Thus extremely different countable order types can have the same membership number. This is a mathematical limitation of the invariant, not a limitation of the ordinal notation.

The countable scattered classification also covers every countable compact Hausdorff space. For completeness, the fact that such a space is scattered is proved in section A, rather than invoked as an unproved classification of countable compact spaces by ordinals.

11 A purely order-theoretic form for finite posets

Equip a finite poset P with the Alexandrov topology of upper sets. This is T_0 , and it is Hausdorff only when the order is an antichain. It is important not to mistake this topology for the order topology on the infinite ordinal spaces studied above.

Fix a target $A \subseteq P$ and color points by their membership in A . A strictly increasing chain is *alternating* if consecutive colors differ. Let a_b be the largest number of elements in an alternating chain whose greatest element has color b , with $a_b = 0$ if no point has that color.

Theorem 11.1 (Alternating-chain formula). *For every nonempty finite poset P and target A ,*

$$N_0(A, P) = 1 + a_1, \quad N_1(A, P) = 1 + a_0, \quad (11.1)$$

and therefore

$$\boxed{\text{sm}(A, P) = \lceil \log_2(1 + \min(a_0, a_1)) \rceil.} \quad (11.2)$$

Proof. Closure in the upper-set topology is downward closure. We claim that $F_n^c \neq \emptyset$ exactly when there is an alternating chain of length n ending in color $1 - c$.

More precisely, F_n^c is the downward closure of the least elements of all such chains. This is immediate for $n = 1$. In the next step, intersect with color $(c + n + 1) \bmod 2$ and take downward closure. A point of that color below the least element of an existing length- n alternating chain has the opposite color from that least element, so the inequality is strict and it can be prepended to the chain. Conversely, the first point of any length- $(n + 1)$ chain is produced this way. The induction follows.

The largest index of a nonempty F_n^c is consequently a_{1-c} , giving (11.1). Now apply theorems 4.2 and 5.2. $\square$

Corollary 11.2 (Quadratic-time computation). *Given the comparability relation of an n -element poset and a target, its optimal finite query depth is computable in $O(n^2)$ time after choosing a linear extension.*

Proof. Process points in a linear extension. Set

$$a(x) = 1 + \max\{a(y) : y < x, y \text{ has the opposite color to } x\},$$

where the maximum of the empty set is 0. Then $a_b = \max\{a(x) : x \text{ has color } b\}$. Each comparable pair is inspected at most once. Formula (11.2) completes the computation. $\square$

Example 11.3 (A finite orientation conflict). An alternating four-element chain has profile $(4, 5)$ or $(5, 4)$, depending on its orientation. Its target takes two questions. Two disjoint chains with opposite orientations have profile $(5, 5)$, so their combined target takes three questions. The archive contains an explicit optimal query tree for this eight-point example.

Remark 11.4. This order-theoretic formula is not obtained by applying the Hausdorff finite-height theorem to a non-Hausdorff space. Both results are separate consequences of the general finite normal form, and their hypotheses are kept distinct.

12 Algorithms, exact checks, and reproducibility

The proof of theorem 4.2 gives a strategy compiler. First construct the two canonical closure sequences; choose the shorter one; then binary-search its layer indices. At a node where the possible indices are $a, \dots, b - 1$, choose $j = \lfloor (a + b)/2 \rfloor$ and ask $X \setminus F_j$. A yes answer selects $a, \dots, j - 1$, and a no answer selects $j, \dots, b - 1$.

For a general infinite space, computing a closure is a mathematical operation, not necessarily an effective algorithm on a presentation of the space. The complexity claim in section 11 is specifically for finite posets whose comparability relation is supplied.

12.1 Independent verification paths

The program `code/verify.py` uses only the Python standard library. For each finite poset and every target, it compares three independently implemented calculations: exhaustive minimax over all upper-set questions, canonical downward-closure profiles, and the alternating-chain dynamic program. It also constructs a balanced strategy tree from the shorter closure chain and checks the color returned on every point.

The exhaustive minimax recurrence is

$$d_A(S) = \begin{cases} 0, & S \text{ is monochromatic,} \\ 1 + \min_U \max\{d_A(S \cap U), d_A(S \setminus U)\}, & \text{otherwise,} \end{cases}$$

where U ranges over upper sets splitting S nontrivially. States are encoded as bit masks, and the recursion is memoized. This recurrence does not use the closure formula being tested.

Number of points	Naturally labelled posets	Targets checked
1	1	2
2	2	8
3	7	56
4	40	640
5	357	11,424
6	4,824	308,736
Total	5,231	320,866

All checks passed. “Naturally labelled” means that every strict comparison $i <_P j$ satisfies $i < j$ as an integer. These are not counts of isomorphism classes. Every finite poset has at least one such labelling, so the search covers every isomorphism type through six points, with repetitions possible.

The included run took approximately 13 seconds in the execution environment used for this report. That observation is not a portable performance guarantee. The machine-readable result is in `data/verification.json`; the text log, count table, ordinal-value table, sharp-family table, and eight-point strategy certificate are provided alongside it.

12.2 Scope of the computational evidence

Finite upper-set topologies are not convergent sequences and do not model the Hausdorff accumulation behavior of ordinal intervals. Accordingly, the finite checks do *not* establish the infinite-space counterexample, the Cantor–Bendixson formula, or its transfinite consequences. Those results follow from the proofs in sections 2, 6 and 10. The computation instead checks the general finite game/closure/alternation mechanism in a large family where a direct independent minimax computation is possible.

To reproduce the included exhaustive run, execute

```
python3 code/verify.py --max-n 6
```

To rebuild the PDF, execute `make pdf`, or run `pdflatex` three times on `article.tex`. All inputs needed for compilation are local; no network access, external bibliography processor, or data regeneration is required.

13 What is settled and what is not

The topological-sum identity (1.1) is false as stated. The counterexample can be expressed using the ordinary ordinal intervals $[0, \omega]$ and $[0, \omega \cdot 2]$, and the one-move defect is sharp at every positive finite level. The proof also yields exact computations for every Hausdorff space of finite Cantor–Bendixson height and every countable scattered Hausdorff space.

The two central distinctions are structural. First, the number of alternating layers is different from the number of adaptive questions: they are related by a ceiling logarithm. Second, the minimum over the two orientations is different from the pair of oriented lengths. Taking a disjoint union can force a common orientation and increase the minimum length by one; when that passes a power of two, it costs a full additional query.

No transfinite version of (4.4) is asserted. In particular, the symbol ∞ in the finite profile is not an ordinal rank and cannot simply be inserted into an ordinal logarithm. Nothing here determines the games at arbitrary uncountable lengths or resolves the other problems posed in [1]. Those are separate questions.

The research claim should be read at its demonstrated scale: an explicit negative solution to an explicitly stated question, together with a complete additional classification on a natural order-topological class. The proofs are intended to be independently checkable; their priority and presentation can be assessed separately from their mathematical content.

A Why countable compact Hausdorff spaces are scattered

This appendix supplies the fact used after corollary 10.3 without requiring the full ordinal classification of countable compact spaces.

Lemma A.1. *A nonempty compact Hausdorff space with no isolated points has at least $2^{\aleph_0}$ points.*

Proof. Construct nonempty open sets U_s indexed by finite binary strings. Start with the whole space. Inside each U_s , select two distinct points; this is possible because no nonempty open set is a singleton. By regularity, choose nonempty open neighborhoods U_{s0}, U_{s1} with disjoint closures contained in U_s .

For each infinite binary string z , the compact sets $\overline{U_{z|n}}$ are nonempty and nested, so their intersection is nonempty. Choose one point from each intersection. Distinct binary strings split at some level and hence give points in disjoint closed sets. This produces an injection of $2^{\mathbb{N}}$ into the space. $\square$

Proposition A.2. *Every countable compact Hausdorff space is scattered.*

Proof. If a space is not scattered, some nonempty subspace B has no isolated points. Its closure F has no isolated points either: an isolated point of F would, by density, belong to B and be isolated in B . In a compact Hausdorff space, F is compact Hausdorff. The preceding lemma would make F uncountable, a contradiction. $\square$

This proof also shows why no additional set-theoretic hypothesis such as the continuum hypothesis is needed. Uncountability of $2^{\mathbb{N}}$ is sufficient.

B A proof audit and implementation checklist

The following points isolate possible sources of off-by-one or topology errors.

Zero questions. A constant target has $L = 1$, not $L = 0$, on a nonempty space. Thus its depth is $\lceil \log_2 1 \rceil = 0$. The empty space is assigned $\text{sm} = 0$ separately.

Derivative rank. The integer h is the index of the *last nonempty* derivative, not the height. The height is $h + 1$. A convergent sequence has $h = 1$ and height 2.

First closure. The general estimate is $F_n \subseteq X^{(n-1)}$ for $n \geq 1$, not $F_n \subseteq X^{(n)}$. The extra first step is exactly what creates the two oriented lengths $h + 1$ and $h + 2$ for a parity target.

Hausdorff separation. The lower bound with two terminal points uses disjoint open neighborhoods. The finite-height theorem is therefore stated for Hausdorff spaces. Its proof is not silently applied to arbitrary T_0 posets.

Arbitrary answers. The binary-search proof wins on every answer sequence, including one whose state becomes empty. In the ω -length certification strategy, an all-no certification path has empty final intersection. There is no hidden assumption that the Hider fixes a point beforehand.

Infinite sums. In the oriented finite-profile formula, an unbounded supremum of integers is recorded as ∞ , meaning “no finite decomposition”. The separate certification argument is needed to obtain the actual ordinal value ω .

Independent checking. The minimax implementation enumerates all open queries. It does not import the closed-layer answer as its return value. The mathematical proof and the finite computational check are deliberately distinct.

References

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